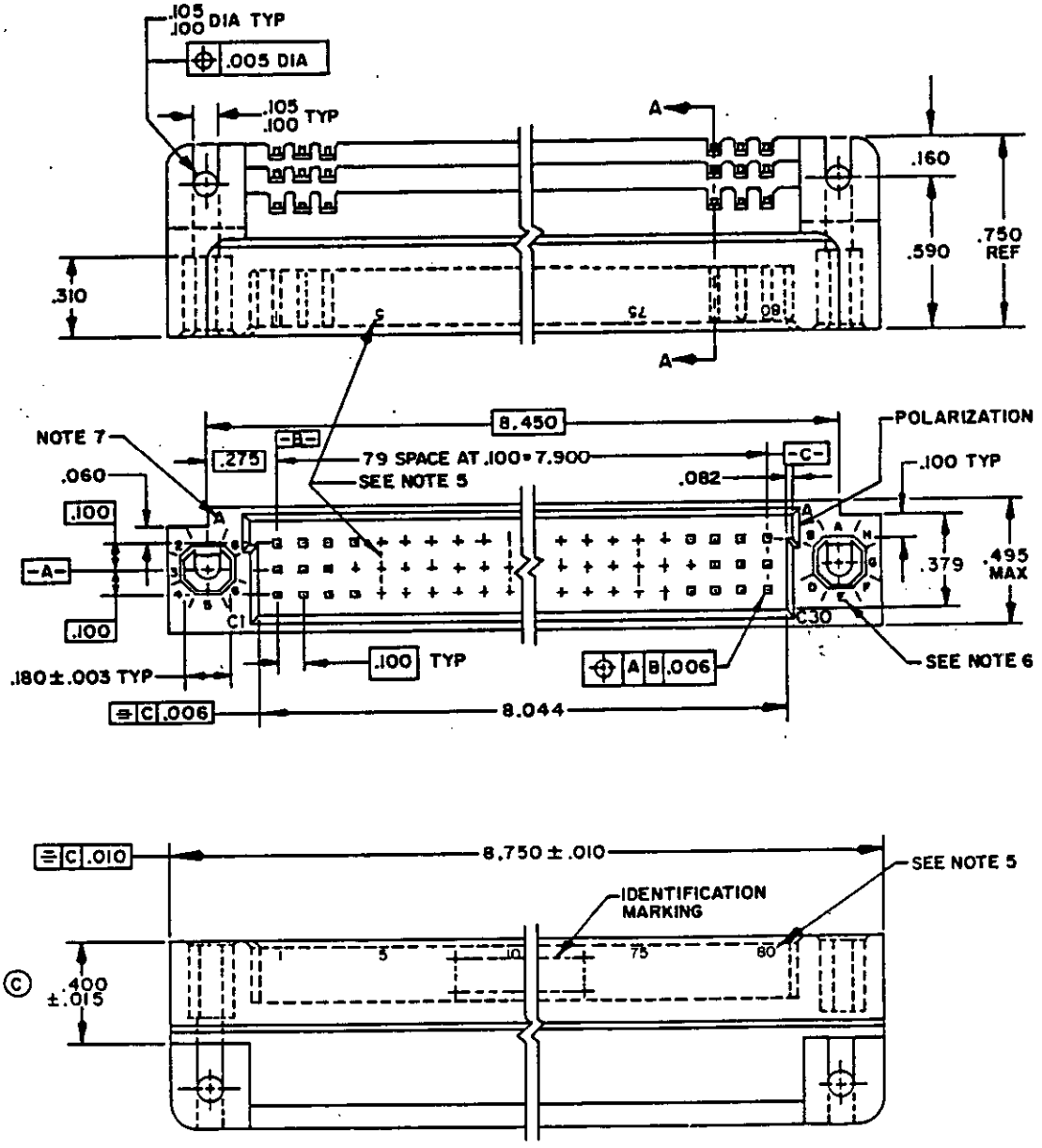
MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: PLUG, RIGHT ANGLE, 90 THROUGH 240 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.100 INCH SPACING)

This specification sheet is approved for use by Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

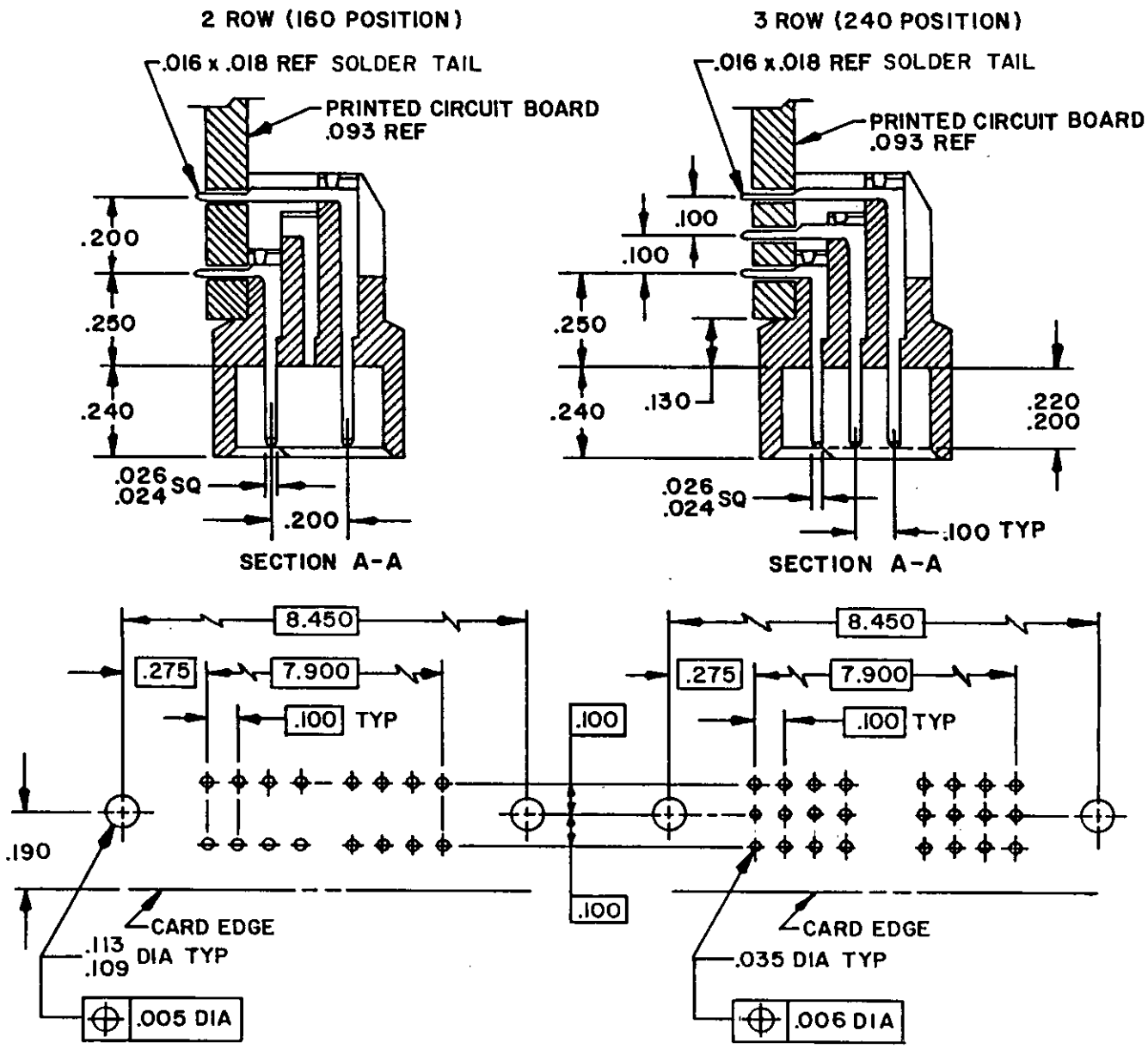
The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-55302.



Ⓒ FIGURE 1. Connectors, kit plug (.100 spacing) 160 and 240 contacts.

Ⓒ denotes change

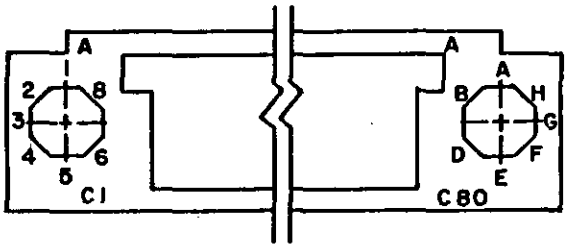
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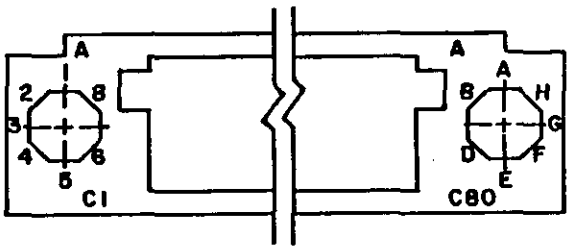
Recommended circuit layouts

© FIGURE 1. Connectors, kit plug (.100 spacing) 160 and 240 contacts - Continued.

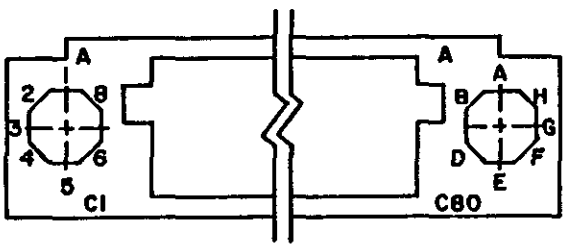
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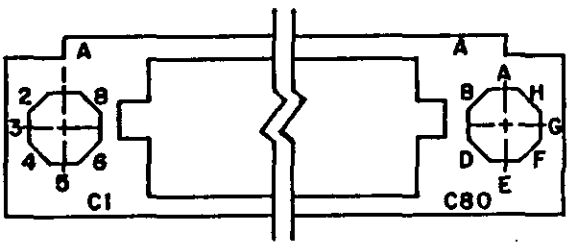
TYPE I



TYPE II



TYPE III



TYPE IV

POLARIZATION

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified; tolerance is ±.005 (0.13 mm) on decimals and +2° on angles.
4. These connectors mate with connectors specified in MIL-C-55302/130.
5. Numbers indicating end cavities stamped both sides. Lines indicating every five cavities stamped on mating face, except end cavities.
6. Location indicators embossed on surface.
7. Letters indicating rows and numbers indicating end cavities stamped this surface.

Inches	mm	Inches	mm	Inches	mm
.003	0.08	.109	2.76	.310	7.87
.005	0.13	.113	2.87	.377	9.58
.006	0.15	.130	3.30	.400	10.16
.010	0.25	.160	4.06	.495	12.57
.024	0.64	.180	4.57	.590	14.99
.026	0.66	.200	5.08	.750	19.05
.035	0.89	.220	5.59	7.900	200.66
.060	1.52	.240	6.10	8.044	204.32
.082	2.08	.250	6.35	8.450	214.63
.100	2.54	.275	6.99	8.750	222.25
.105	2.67				

© FIGURE 1. Connectors, kit plug (.100 spacing) 160 and 240 contacts - Continued.

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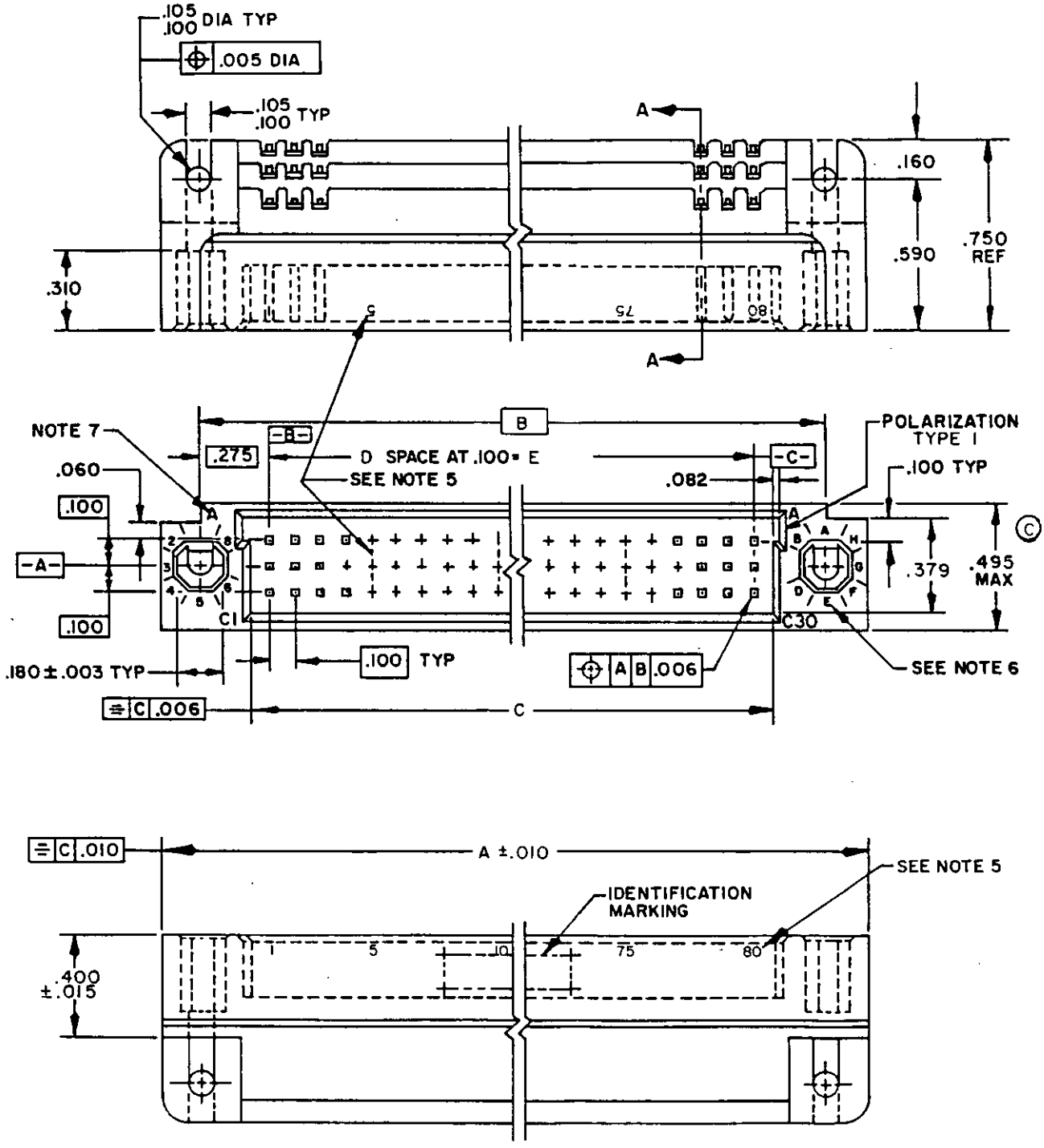
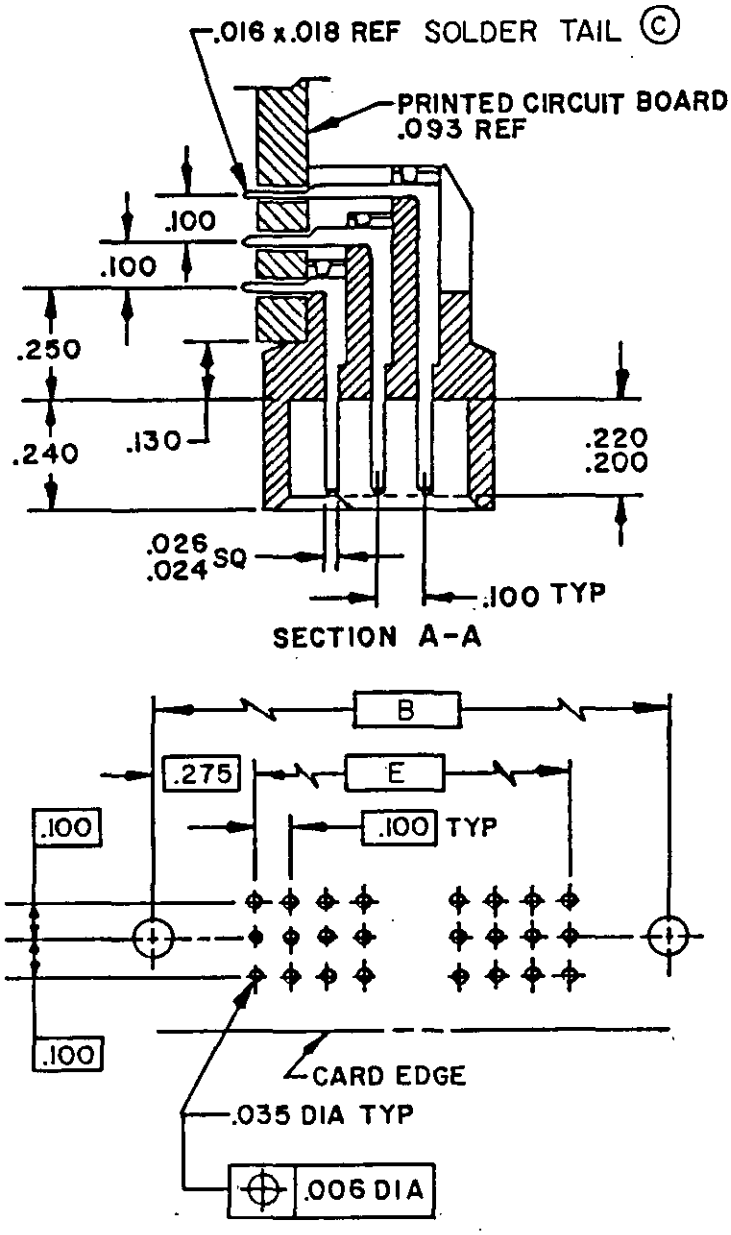


FIGURE 2. Connectors, plug (.100 spacing) 90 through 240 contact positions.

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Inches	mm	
.003	.08	
.005	0.13	
.006	0.15	
.010	0.25	
.024	0.64	
.026	0.66	
.035	0.89	
.060	1.52	
.082	2.08	
.100	2.54	
.105	2.67	
.109	2.76	
.113	2.87	
.130	3.30	
.160	4.06	
.180	4.57	
.200	5.08	
.220	5.59	
.240	6.10	
.250	6.35	
.275	6.99	
.310	7.87	
.377	9.58	
.400	10.16	©
.495	12.57	
.590	14.99	
.750	19.05	
7.900	200.66	
8.044	204.32	
8.450	214.63	
8.750	222.25	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm) on decimals and +2° on angles.
4. These connectors mate with connectors specified in MIL-C-55302/130.
5. Numbers indicating end cavities stamped both sides. Lines indicating every five cavities stamped on mating face, except end cavities.
6. Location indicators embossed on surface.
7. Letters indicating rows and numbers indicating end cavities stamped this surface.

FIGURE 2. Connectors, plug (.100 spacing) 90 through 240 contact positions - Continued.

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TABLE I. Dash numbers, polarization types, and contact types for 160 and 240 contact positions.

Dash number	Number contacts	Type polarization
01	160	I
02	160	II
03	160	III
04	160	IV
05	240	I
06	240	II
07	240	III
08	240	IV

TABLE II. Dash numbers and dimensions.
(For 90 through 240 contact positions)

Dash no.	No. of contacts	A	B	C	D	E	Contact ident. no.
09	90	3.750 (95.25)	3.450 (87.63)	3.044 (77.32)	29	2.900 (73.66)	1-30
10	105	4.250 (107.95)	3.950 (100.33)	3.544 (90.02)	34	3.400 (86.36)	1-35
11	120	4.750 (120.65)	4.450 (113.03)	4.044 (102.72)	39	3.900 (99.06)	1-40
12	135	5.250 (133.35)	4.950 (125.73)	4.544 (115.42)	44	4.400 (111.76)	1-45
13	150	5.750 (146.05)	5.450 (138.43)	5.044 (128.12)	49	4.900 (124.46)	1-50
14	165	6.250 (158.75)	5.950 (151.13)	5.544 (140.82)	54	5.400 (137.16)	1-55
15	180	6.750 (171.45)	6.450 (163.83)	6.044 (153.52)	59	5.900 (149.86)	1-60
16	195	7.250 (184.15)	6.950 (176.53)	6.544 (166.22)	64	6.400 (162.56)	1-65
17	210	7.750 (196.85)	7.450 (189.23)	7.044 (178.92)	69	6.900 (175.26)	1-70
18	240	8.750 (222.25)	8.450 (214.63)	8.044 (204.32)	79	7.900 (200.66)	1-80

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REQUIREMENTS:

Design and construction:

Dimensions and configurations: See figures 1 and 2, and table II.

© Material:

© Body: In accordance with MIL-C-55302.

© Plating:

Contact engagement area (.150 inch minimum length): Gold over nickel per MIL-C-55302.

Solder tail area (.160 inch minimum length): Tin lead plate over nickel per MIL-C-55302.

Contact identification: See figures 1 and 2.

© Current rating: 3.0 amperes, maximum per contact; 2.25 amperes continuous per contact at 75°F.

Keying: See MIL-C-55302/31. Two keys and two key retaining rivets packaged with connectors for -01 through -08 only. For -09 through -18, keys and rivets shall be ordered separately.

Mating and unmating: The maximum mating force in pounds shall be the number of contacts multiplied by 0.25; the withdrawal force in pounds shall be a minimum of 0.025 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance: No individual contact pair shall have a resistance exceeding 0.020 ohm.

Contact retention: 3 pounds, minimum.

Dielectric withstanding voltage:

Sea level: 900 volts rms.
High altitude: 200 volts rms.

Part number: M55302/129- (and dash number from table I for 160 and 240 contact positions).

M55302/129- (and dash numbers from table II for 90 through 240 contact positions).

© A customer-supplied teflon cover is recommended to prevent heat damage to the connector during soldering operations.

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Custodian:
Navy - EC

Preparing activity:
Navy - EC

Agent:
DLA - ES

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